

Effect of lemon verbena supplementation on muscular damage markers, proinflammatory cytokines release and neutrophils' oxidative stress in chronic ...

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Intense exercise is directly related to muscular damage and oxidative stress due to excessive reactive oxygen species (ROS) in both, plasma and white blood cells. Nevertheless, exercise-derived ROS are essential to regulate cellular adaptation to exercise. Studies on antioxidant supplements have provided controversial results. The purpose of this study was to determine the effect of moderate antioxidant supplementation (lemon verbena extract) in healthy male volunteers that followed a 90-min running eccentric exercise protocol for 21 days. Antioxidant enzymes activities and oxidative stress markers were measured in neutrophils. Besides, inflammatory cytokines and muscular damage were determined in whole blood and serum samples, respectively. Intense running exercise for 21 days induced antioxidant response in neutrophils of trained male through the increase of the antioxidant enzymes catalase, glutathione peroxidase and glutathione reductase. Supplementation with moderate levels of an antioxidant lemon verbena extract did not block this cellular adaptive response and also reduced exercise-induced oxidative damage of proteins and lipids in neutrophils and decreased myeloperoxidase activity. Moreover, lemon verbena supplementation maintained or decreased the level of serum transaminases activity indicating a protection of muscular tissue. Exercise induced a decrease of interleukin-6 and interleukin-1 β levels after 21 days measured in basal conditions, which was not inhibited by antioxidant supplementation. Therefore, moderate antioxidant supplementation with lemon verbena extract protects neutrophils against oxidative damage, decreases the signs of muscular damage in chronic running exercise without blocking the cellular adaptation to exercise.